Expert Systems in Artificial Intelligence

Simulating Human Expertise Through Logic & Reasoning

What is an Expert System?

- Definition:
  An **Expert System** is a computer program that simulates the decision-making
- ability of a human expert.
- Purpose:
  To solve complex problems by reasoning through knowledge.

Characteristics of Expert Systems

- High performance
- Understandable (explanation capability)
- Reliable and consistent
- Domain-specific expertise
- Can deal with uncertainty (using confidence levels or fuzzy logic)

Components of an Expert System

- **Knowledge Base**
  Domain-specific facts & heuristics
- **Inference Engine**
  Logical rules to apply the knowledge
- **User Interface**
  Allows interaction between user and system
- **Explanation Facility**
  Explains reasoning behind decisions
- **Knowledge Acquisition Module**
  Helps update the knowledge base

Architecture Diagram

- Visual flowchart showing:
- User → UI → Inference Engine ↔ Knowledge Base
- Explanation and Knowledge Acquisition modules connecting to the system

Working of an Expert System

- Step-by-step process:

1. User enters query or problem
2. Inference engine searches knowledge base
3. Applies logical rules
4. Provides solution and explanation

Types of Inference Techniques

- **Forward Chaining**
- Data-driven: Starts with known facts
- **Backward Chaining**
- Goal-driven: Starts with a hypothesis
- Example: Diagnosing diseases using symptoms (forward), testing a diagnosis (backward)

Knowledge Representation

- Methods used in Expert Systems:
- **Rules (IF-THEN statements)**
- **Semantic Networks**
- **Frames**
- **Ontologies**
- Example rule:
  IF fever AND cough THEN flu

Knowledge Acquisition

- Ways to build the knowledge base:
  - Interviews with human experts
  - Case studies
  - Text mining
  - Expert surveys
- Challenge: Extracting tacit knowledge

Example Expert Systems

Name	Domain	Description
MYCIN	Medical	Diagnosed bacterial infections
DENDRAL	Chemistry	Analyzed mass spectrometry data
CLIPS	General-purpose	NASA-developed expert system shell
XCON	Business	Configured computer systems

Case Study – MYCIN

- Medical expert system for diagnosing infections
- Used backward chaining
- Included explanation facility
- Performance rivaled human doctors in tests

Applications of Expert Systems

- Medical diagnosis
- Mineral exploration
- Financial decision-making
- Troubleshooting machinery
- Agricultural advisory
- Cybersecurity threat detection

Limitations of Expert Systems

- Combining with:
- Machine Learning (for learning new rules)
- Natural Language Processing (for better interfaces)
- Knowledge Graphs
- Cloud and IoT for real-time expert systems

Tools for Building Expert Systems

- **CLIPS** (C Language Integrated Production System)
- **Prolog**
- **Jess** (Java Expert System Shell)
- **Drools** (Business rule engine for Java)

Future of Expert Systems

- Neuro-symbolic systems
- Explainable AI (XAI)
- Domain-adaptive expert systems
- Real-time diagnostics with IoT sensors
- Embedded expert systems in robotics